

Trading Algorithm

APCS | CS408: Computation Finance
Group Name: 02

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Checklist:

☒ Position Entry Point
☒ Target Market

☒ Take-Profit Point
☒ Stop-Loss Point

☒ Position Sizing
☒ Trading Tactics

Hypothesis

Bollinger Bands is a technical analysis tool used to measure price volatility and identify key points in the stock market. It assumes that stock prices typically fluctuate around a mean value, which can be measured by standard deviation.

This approach mainly focuses on the upper and lower bands, which are calculated based on the standard deviation of prices from the moving average. Typically placed at a standard deviation (usually 2) away from the moving average, the bands represent potential price levels that the price may fluctuate within.

When the stock price approaches the upper band, it may indicate a potential selling point as the price is assumed to be overvalued. Conversely, when the price approaches the lower band, it may suggest a potential buying point as the price is assumed to be undervalued.

Target Market

VN30 Index Future Contract – VN30F1M

Strategy Group

Momentum
☒ Mean-Reversion

Grid
Scalping

Market-Making

When to Open Positions

(Position Entry Point + Position Sizing + Trading Tactics)

- Independent variable: VN30F1M price by ticks.
- Dependent variable:
 - $\text{MiddleBand}_t = \text{SMA}(n)_t = (\text{TP}_t + \text{TP}_{(t-1)} + \text{TP}_{(t-2)} + \dots + \text{TP}_{(t-m)}) \div m$
 - $\text{UpperBand}_t = \text{SMA}(n)_t + m * \sigma(\text{SMA}(n)_t)$
 - $\text{LowerBand}_t = \text{SMA}(n)_t - m * \sigma(\text{SMA}(n)_t)$

Where:

TP_t is the typical price of VN30F1M at day t, $\text{TP}_t = (\text{High}_t + \text{Low}_t + \text{Close}_t) / 3$

n is the number of days in the smoothing period (typically 10).

m is the number of standard deviations (typically 1.9).

- Place a limit order (LO) to buy at the ceiling price or sell at the floor price as soon as there's a signal to open a position.
 - If $\text{TP}_{t-1} > \text{LowerBand}_{t-1}$ and $\text{TP}_t \leq \text{LowerBand}_t$, then open a Buy position;
 - If $\text{TP}_{t-1} < \text{UpperBand}_{t-1}$ and $\text{TP}_t \geq \text{UpperBand}_t$, then open a Sell position.
- Trade 01 contract for each position, and open only 1 position at a time.

When to Close Positions

(Take-Profit Point/Stop-Loss Point + Position Sizing + Trading Tactics)

- **Take Profit:**
 - If the price reaches the upper band when having a Buy position, place an order to sell at the floor price to close the position.
 - If the price reaches the lower band when having a Sell position, place an order to buy at the ceiling price to close the position.
- **Stop Loss:**
 - Set threshold = $0.5 * \sigma(\text{SMA}(n)_t)$ as the threshold for stop loss action
 - If $\text{TP}_t \geq \text{UpperBand}_t + \text{threshold}$ when having a Sell position then place an order to buy at the ceiling price to close the position.
 - If $\text{TP}_t \leq \text{LowerBand}_t - \text{threshold}$ when having a Buy position then place an order to sell at the floor price to close the position.
- Positions can remain overnight as SMA is being calculated on a daily level.
- Trade 01 contract for each position, and close only 1 position at a time.
- The process takes place at the beginning of the contract. If none of the aforementioned closing conditions is met after the duration of the contract, then place an order to buy at the floor/ceiling price to close the Sell/Buy position.